General Aptitude

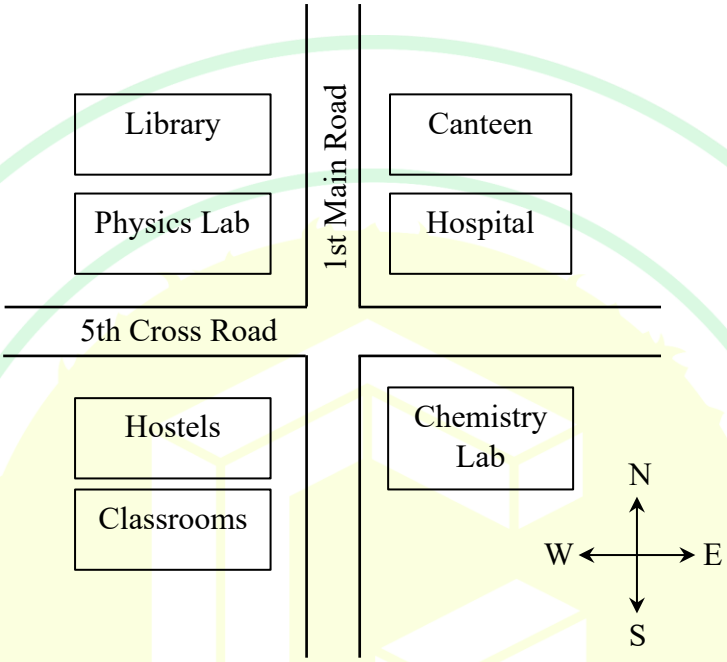
Q.1 – Q.5 Carry ONE mark Each

Q.1	Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select the option with the correct sequence of articles to fill in the blanks.
(A)	a; a; an
(B)	the; an; a
(C)	a; an; a
(D)	an; an; a

Q.2	The CEO's decision to downsize the workforce was considered myopic because it sacrificed long-term stability to accommodate short-term gains. Select the most appropriate option that can replace the word "myopic" without changing the meaning of the sentence.
(A)	visionary
(B)	shortsighted
(C)	progressive
(D)	innovative

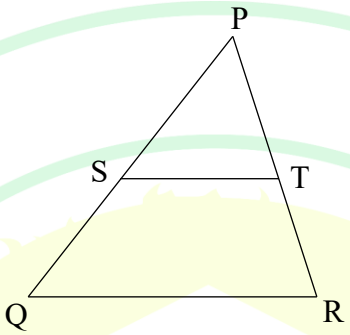
Q.3	The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4. How many students does the class have?
(A)	25
(B)	28
(C)	30
(D)	32

Q.4	Consider the relationships among P, Q, R, S, and T: • P is the brother of Q. • S is the daughter of Q. • T is the sister of S. • R is the mother of Q. The following statements are made based on the relationships given above. (1) R is the grandmother of S. (2) P is the uncle of S and T. (3) R has only one son. (4) Q has only one daughter. Which one of the following options is correct?
(A)	Both (1) and (2) are true.
(B)	Both (1) and (3) are true.
(C)	Only (3) is true.
(D)	Only (4) is true.

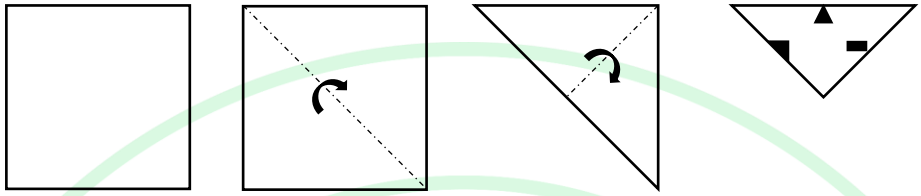
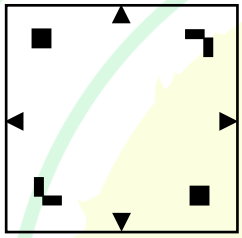
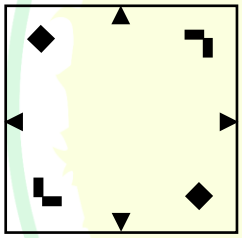
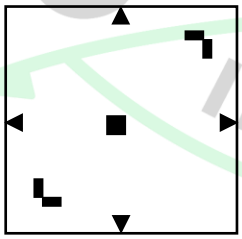
Q.5	According to the map shown in the figure, which one of the following statements is correct? Note: The figure shown is representative.
	
(A)	The library is located to the northwest of the canteen.
(B)	The hospital is located to the east of the chemistry lab.
(C)	The chemistry lab is to the southeast of physics lab.
(D)	The classrooms and canteen are next to each other.

Q.6 – Q.10 Carry TWO marks Each

Q.6	"I put the brown paper in my pocket along with the chalks, and possibly other things. I suppose every one must have reflected how primeval and how poetical are the things that one carries in one's pocket: the pocket-knife, for instance the type of all human tools, the infant of the sword. Once I planned to write a book of poems entirely about the things in my pocket. But I found it would be too long: and the age of the great epics is past." (From G.K. Chesterton's "A Piece of Chalk") Based only on the information provided in the above passage, which one of the following statements is true?
(A)	The author of the passage carries a mirror in his pocket to reflect upon things.
(B)	The author of the passage had decided to write a poem on epics.
(C)	The pocket-knife is described as the infant of the sword.
(D)	Epics are described as too inconvenient to write.

Q.7	In the diagram, the lines QR and ST are parallel to each other. The shortest distance between these two lines is half the shortest distance between the point P and line QR. What is the ratio of the area of the triangle PST to the area of the trapezium SQRT? Note: The figure shown is representative.
	
(A)	$\frac{1}{3}$
(B)	$\frac{1}{4}$
(C)	$\frac{2}{5}$
(D)	$\frac{1}{2}$

Q.8	A fair six-faced dice, with the faces labelled ‘1’, ‘2’, ‘3’, ‘4’, ‘5’, and ‘6’, is rolled thrice. What is the probability of rolling ‘6’ exactly once?
(A)	$\frac{75}{216}$
(B)	$\frac{1}{6}$
(C)	$\frac{1}{18}$
(D)	$\frac{25}{216}$

Q.9	A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few cuts are made as shown in figure (IV). Which one of the following patterns will be obtained when the paper is unfolded? Note: The figures shown are representative.
	 (I) (II) (III) (IV)
(A)	
(B)	
(C)	
(D)	

Q.10	A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. **The order in which the scoops are purchased is inconsequential.** If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase?
(A)	4
(B)	20
(C)	24
(D)	48

Q.11 – Q.35 Carry ONE mark Each

Q.11	Suppose a program is running on a non-pipelined single processor computer system. The computer is connected to an external device that can interrupt the processor asynchronously. The processor needs to execute the interrupt service routine (ISR) to serve this interrupt. The following steps (not necessarily in order) are taken by the processor when the interrupt arrives: (i) The processor saves the content of the program counter. (ii) The program counter is loaded with the start address of the ISR. (iii) The processor finishes the present instruction. Which ONE of the following is the CORRECT sequence of steps?
(A)	(iii), (i), (ii)
(B)	(i), (iii), (ii)

(C)	(i), (ii), (iii)
(D)	(iii), (ii), (i)

Q.12	Which ONE of the following statements is **FALSE** regarding the symbol table?
(A)	Symbol table is responsible for keeping track of the scope of variables.
(B)	Symbol table can be implemented using a binary search tree.
(C)	Symbol table is not required after the parsing phase.
(D)	Symbol table is created during the lexical analysis phase.

Q.13	Which ONE of the following techniques used in compiler code optimization uses live variable analysis?
(A)	Run-time function call management
(B)	Register assignment to variables
(C)	Strength reduction
(D)	Constant folding

Q.14	Consider a demand paging memory management system with 32-bit logical address, 20-bit physical address, and page size of 2048 bytes. Assuming that the memory is byte addressable, what is the maximum number of entries in the page table?
(A)	2^{21}
(B)	2^{20}
(C)	2^{22}
(D)	2^{24}

Q.15	A schedule of three database transactions T_1, T_2, and T_3 is shown. $R_i(A)$ and $W_i(A)$ denote read and write of data item A by transaction $T_i, i = 1,2,3$. The transaction T_1 aborts at the end. Which other transaction(s) will be required to be rolled back?
	$R_1(X)\ W_1(Y)\ R_2(X)\ R_2(Y)\ R_3(Y)\ ABORT(T_1)$
(A)	Only T_2
(B)	Only T_3
(C)	Both T_2 and T_3
(D)	Neither T_2 nor T_3

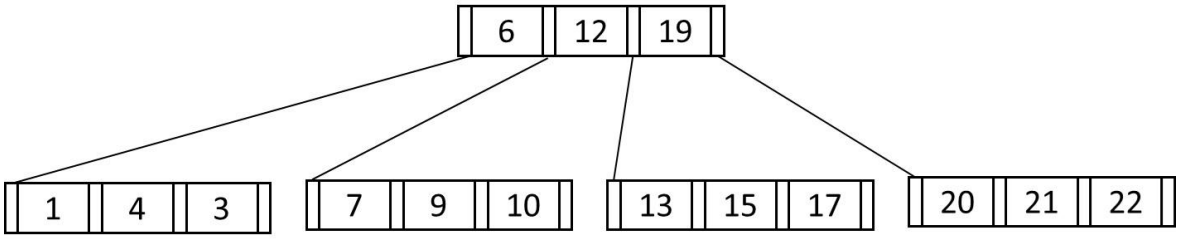
<table>
<tr><td>Q.16</td><td>Identify the ONE CORRECT matching between the OSI layers and their corresponding functionalities as shown.</td></tr>
<tr><td></td><td>
<table>
<tr><th><u>OSI Layers</u></th><th><u>Functionalities</u></th></tr>
<tr><td>(a) Network layer</td><td>(I) Packet routing</td></tr>
<tr><td>(b) Transport layer</td><td>(II) Framing and error handling</td></tr>
<tr><td>(c) Datalink layer</td><td>(III) Host to host communication</td></tr>
</table>
</td></tr>
<tr><td>(A)</td><td>(a)-(I), (b)-(II), (c)-(III)</td></tr>
<tr><td>(B)</td><td>(a)-(I), (b)-(III), (c)-(II)</td></tr>
<tr><td>(C)</td><td>(a)-(II), (b)-(I), (c)-(III)</td></tr>
<tr><td>(D)</td><td>(a)-(III), (b)-(II), (c)-(I)</td></tr>
</table>

Q.17	$g(.)$ is a function from A to B, $f(.)$ is a function from B to C, and their composition defined as $f(g(.))$ is a mapping from A to C. If $f(.)$ and $f(g(.))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(.)$?
(A)	$g(.)$ must be an onto (surjective) function.
(B)	$g(.)$ must be a one-to-one (injective) function.
(C)	$g(.)$ must be a bijective function, that is, both one-to-one and onto.
(D)	$g(.)$ is not required to be a one-to-one or onto function.

Q.18	Let G be any undirected graph with positive edge weights, and T be a minimum spanning tree of G. For any two vertices, u and v, let $d_1(u,v)$ and $d_2(u,v)$ be the shortest distances between u and v in G and T, respectively. Which ONE of the options is CORRECT for all possible G, T, u and v?
(A)	$d_1(u,v) = d_2(u,v)$
(B)	$d_1(u,v) \leq d_2(u,v)$
(C)	$d_1(u,v) \geq d_2(u,v)$
(D)	$d_1(u,v) \neq d_2(u,v)$

Q.19	Consider the following context-free grammar G, where S, A, and B are the variables (non-terminals), a and b are the terminal symbols, S is the start variable, and the rules of G are described as: $S \rightarrow aaB \mid Abb$ $A \rightarrow a \mid aA$ $B \rightarrow b \mid bB$ Which ONE of the languages $L(G)$ is accepted by G?
(A)	$L(G) = \{a^2 b^n \mid n \geq 1\} \cup \{a^n b^2 \mid n \geq 1\}$
(B)	$L(G) = \{a^n b^{2n} \mid n \geq 1\} \cup \{a^{2n} b^n \mid n \geq 1\}$
(C)	$L(G) = \{a^n b^n \mid n \geq 1\}$
(D)	$L(G) = \{a^{2n} b^{2n} \mid n \geq 1\}$

Q.20	Consider the following recurrence relation: $T(n) = 2T(n-1) + n2^n$ for $n > 0, \quad T(0) = 1.$ Which ONE of the following options is CORRECT?
(A)	$T(n) = \Theta(n^2 2^n)$
(B)	$T(n) = \Theta(n2^n)$
(C)	$T(n) = \Theta((\log n)^2\, 2^n)$
(D)	$T(n) = \Theta(4^n)$

Q.21	Consider the following B^+ tree with 5 nodes, in which a node can store at most 3 key values. The value 23 is now inserted in the B^+ tree. Which of the following options(s) is/are CORRECT?
	
(A)	None of the nodes will split.
(B)	At least one node will split and redistribute.
(C)	The total number of nodes will remain same.
(D)	The height of the tree will increase.

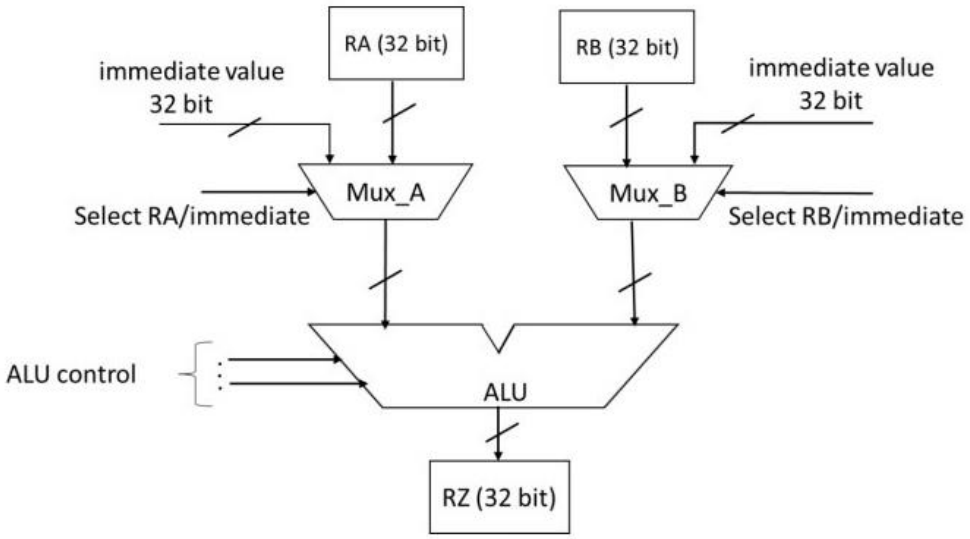
Q.22	Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2, and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?
(A)	P3: SYN = 1, ACK = 1
(B)	P2: SYN = 1, ACK = 1
(C)	P2: SYN = 0, ACK = 1
(D)	P1: SYN = 1

Q.23	Consider the given system of linear equations for variables x and y, where k is a real-valued constant. Which of the following option(s) is/are CORRECT?
	$$x + ky = 1$$ $$kx + y = -1$$
(A)	There is exactly one value of k for which the above system of equations has no solution.
(B)	There exist an infinite number of values of k for which the system of equations has no solution.
(C)	There exists exactly one value of k for which the system of equations has exactly one solution.
(D)	There exists exactly one value of k for which the system of equations has an infinite number of solutions.

Q.24	Let X be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which of the following statement(s) is/are CORRECT, where a, b, c, d, e are Boolean variables?
(A)	$X(a, b, X(c, d, e)) = X(X(a, b, c), d, e)$
(B)	$X(a, b, X(a, b, c)) = X(a, b, c)$
(C)	$X(a, b, X(a, c, d)) = (X(a, b, a)$ AND $X(c, d, c))$
(D)	$X(a, b, c) = X(a, X(a, b, c), X(a, c, c))$

Q.25	The number -6 can be represented as 1010 in 4-bit 2's complement representation. Which of the following is/are CORRECT 2's complement representation(s) of -6?
(A)	1000 1010 in 8-bits
(B)	1111 1010 in 8-bits
(C)	1000 0000 0000 1010 in 16-bits
(D)	1111 1111 1111 1010 in 16-bits

Q.26	Which of the following statement(s) is/are TRUE for any binary search tree (BST) having n distinct integers?
(A)	The maximum length of a path from the root node to any other node is $(n-1)$.
(B)	An inorder traversal will always produce a sorted sequence of elements.
(C)	Finding an element takes $O(\log_2 n)$ time in the worst case.
(D)	Every BST is also a Min-Heap.

Q.27	A partial data path of a processor is given in the figure, where RA, RB, and RZ are 32-bit registers. Which option(s) is/are CORRECT related to arithmetic operations using the data path as shown?
	
(A)	The data path can implement arithmetic operations involving two registers.
(B)	The data path can implement arithmetic operations involving one register and one immediate value.
(C)	The data path can implement arithmetic operations involving two immediate values.
(D)	The data path can only implement arithmetic operations involving one register and one immediate value.

Q.28	A regular language L is accepted by a non-deterministic finite automaton (NFA) with n states. Which of the following statement(s) is/are FALSE?
(A)	L may have an accepting NFA with $< n$ states.
(B)	L may have an accepting DFA with $< n$ states.
(C)	There exists a DFA with $\leq 2^n$ states that accepts L.
(D)	Every DFA that accepts L has $> 2^n$ states.

Q.29	Suppose in a multiprogramming environment, the following C program segment is executed. A process goes into I/O queue whenever an I/O related operation is performed. Assume that there will always be a context switch whenever a process requests for an I/O, and also whenever the process returns from an I/O. The number of times the process will enter the ready queue during its lifetime (not counting the time the process enters the ready queue when it is run initially) is _____. (Answer in integer)

	``
	int main()
	{
	    int x=0,i=0;
	    scanf("%d",&x);
	    for(i=0; i<20; i++)
	    {
	        x = x+20;
	        printf("%d\n",x);
	    }
	    return 0;
	}
	``

Q.30	Let S be the set of all ternary strings defined over the alphabet $\{a, b, c\}$. Consider all strings in S that contain at least one occurrence of two consecutive symbols, that is, “aa”, “bb” or “cc”. The number of such strings of length 5 that are possible is _____. (Answer in integer)

Q.31	Consider the given function $f(x)$. $$f(x) = \begin{cases} ax + b & \text{for } x < 1 \\ x^3 + x^2 + 1 & \text{for } x \geq 1 \end{cases}$$ If the function is differentiable everywhere, the value of b must be _____. (rounded off to one decimal place)

Q.32	A box contains 5 coins: 4 regular coins and 1 fake coin. When a regular coin is tossed, the probability $P(head) = 0.5$ and for a fake coin, $P(head) = 1$. You pick a coin at random and toss it twice, and get two heads. The probability that the coin you have chosen is the fake coin is _____. (rounded off to two decimal places)

Q.33

The pseudocode of a function **fun()** is given below:

```
fun(int A[0,…,n-1]){
    for i=0 to n-2
        for j=0 to n-i-2
            if (A[j]>A[j+1])
                then swap A[j] and A[j+1]

}
```

Let $A[0, \ldots, 29]$ be an array storing 30 distinct integers in descending order. The number of swap operations that will be performed, if the function **fun()** is called with $A[0, \ldots, 29]$ as argument, is _____. (Answer in integer)

Q.34

```
#include <stdio.h>
void foo(int *p, int x){
    *p=x;
}
int main(){
    int *z;
    int a = 20, b = 25;
    z = &a;
    foo(z,b);
    printf("%d",a);
    return 0;
}
```

The output of the given C program is _____. (Answer in integer)

Q.35	The height of any rooted tree is defined as the maximum number of edges in the path from the root node to any leaf node. Suppose a Min-Heap T stores 32 keys. The height of T is _____. (Answer in integer)

Q.36 – Q.65 Carry TWO marks Each

Q.36	Consider a memory system with 1M bytes of main memory and 16K bytes of cache memory. Assume that the processor generates 20-bit memory address, and the cache block size is 16 bytes. If the cache uses direct mapping, how many bits will be required to store all the *tag* values? [Assume memory is byte addressable, $1K=2^{10}$, $1M=2^{20}$.]
(A)	6×2^{10}
(B)	8×2^{10}
(C)	2^{12}
(D)	2^{14}

Q.37	A processor has 64 general-purpose registers and 50 distinct instruction types. An instruction is encoded in 32-bits. What is the maximum number of bits that can be used to store the immediate operand for the given instruction? `ADD R1, #25 // R1 = R1 + 25`
(A)	16
(B)	20
(C)	22
(D)	24

Q.38	A computer has two processors, M_1 and M_2. Four processes P_1, P_2, P_3, P_4 with CPU bursts of 20, 16, 25, and 10 milliseconds, respectively, arrive at the same time and these are the only processes in the system. The scheduler uses non-preemptive priority scheduling, with priorities decided as follows: • M_1 uses priority of execution for the processes as, $P_1 > P_3 > P_2 > P_4$, i.e., P_1 and P_4 have highest and lowest priorities, respectively. • M_2 uses priority of execution for the processes as, $P_2 > P_3 > P_4 > P_1$, i.e., P_2 and P_1 have highest and lowest priorities, respectively. A process P_i is scheduled to a processor M_k, if the processor is free and no other process P_j is waiting with higher priority. At any given point of time, a process can be allocated to any one of the free processors without violating the execution priority rules. Ignore the context switch time. What will be the average waiting time of the processes in milliseconds?
(A)	9.00
(B)	8.75
(C)	6.50
(D)	7.50

Q.39	Consider two relations describing $teams$ and $players$ in a sports league: • $teams(tid, tname)$: tid, $tname$ are team-id and team-name, respectively • $players(pid, pname, tid)$: pid, $pname$, and tid denote player-id, player-name and the team-id of the player, respectively Which ONE of the following tuple relational calculus queries returns the name of the players who play for the team having $tname$ as $'MI'$?
(A)	$\{p.pname \mid p \in players \wedge \exists t\ (t \in teams \wedge p.tid = t.tid \wedge t.tname = 'MI')\}$
(B)	$\{p.pname \mid p \in teams \wedge \exists t\ (t \in players \wedge p.tid = t.tid \wedge t.tname = 'MI')\}$
(C)	$\{p.pname \mid p \in players \wedge \exists t\ (t \in teams \wedge t.tname = 'MI')\}$
(D)	$\{p.pname \mid p \in teams \wedge \exists t\ (t \in players \wedge t.tname = 'MI')\}$

Q.40 A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?

Subnet Address	Subnet Mask (in CIDR notation)	Interface
145.36.0.0	/16	E1
145.36.128.0	/17	E2
145.36.64.0	/18	E3
145.36.255.0	/24	E4
Default	--	E5

(A) E3

(B) E1

(C) E2

(D) E5

Q.41	Let A be a 2×2 matrix as given. $$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$ What are the eigenvalues of the matrix A^{13} ?
(A)	$1,\ -1$
(B)	$2\sqrt{2},\ -2\sqrt{2}$
(C)	$4\sqrt{2},\ -4\sqrt{2}$
(D)	$64\sqrt{2},\ -64\sqrt{2}$

Q.42	Consider the following four variable Boolean function in sum-of-product form $$F(b_3, b_2, b_1, b_0) = \sum(0, 2, 4, 8, 10, 11, 12).$$ where the value of the function is computed by considering $b_3b_2b_1b_0$ as a 4-bit binary number, where b_3 denotes the most significant bit and b_0 denotes the least significant bit. Note that there are no don't care terms. Which ONE of the following options is the CORRECT minimized Boolean expression for F?
(A)	$\bar{b}_1\bar{b}_0 + \bar{b}_2\bar{b}_0 + b_1\,\bar{b}_2b_3$
(B)	$\bar{b}_1\,\bar{b}_0 + \bar{b}_2\,\bar{b}_0$
(C)	$\bar{b}_2\bar{b}_0 + b_1b_2b_3$
(D)	$\bar{b}_0\bar{b}_2 + \bar{b}_3$

Q.43	Let $G(V, E)$ be an undirected and unweighted graph with 100 vertices. Let $d(u, v)$ denote the number of edges in a shortest path between vertices u and v in V. Let the maximum value of $d(u, v)$, $u, v \in V$ such that $u \neq v$, be 30. Let T be any breadth-first-search tree of G. Which ONE of the given options is CORRECT for every such graph G?
(A)	The height of T is exactly 15.
(B)	The height of T is exactly 30.
(C)	The height of T is at least 15.
(D)	The height of T is at least 30.

Q.44	Consider the following two languages over the alphabet $\{a, b\}$: $L_1 = \{ \alpha\beta\alpha \mid \alpha \in \{a,b\}^+ \text{ AND } \beta \in \{a,b\}^+ \}$ $L_2 = \{ \alpha\beta\alpha \mid \alpha \in \{a\}^+ \text{ AND } \beta \in \{a,b\}^+ \}$ Which ONE of the following statements is CORRECT?
(A)	Both L_1 and L_2 are regular languages.
(B)	L_1 is a regular language but L_2 is not a regular language.
(C)	L_1 is not a regular language but L_2 is a regular language.
(D)	Neither L_1 nor L_2 is a regular language.

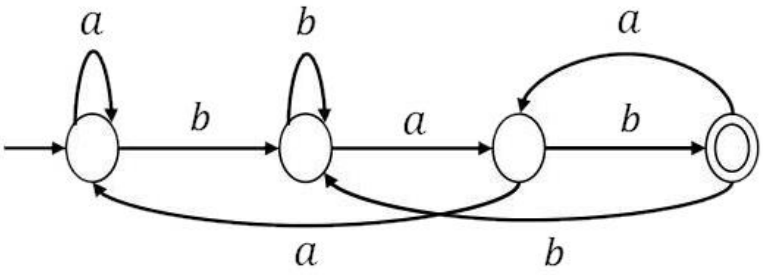
Q.45	Consider the following two languages over the alphabet $\{a, b, c\}$, where m and n are natural numbers. $$L_1 = \{a^m b^m c^{m+n} \mid m, n \geq 1\}$$ $$L_2 = \{a^m b^n c^{m+n} \mid m, n \geq 1\}$$ Which ONE of the following statements is CORRECT?
(A)	Both L_1 and L_2 are context-free languages.
(B)	L_1 is a context-free language but L_2 is not a context-free language.
(C)	L_1 is not a context-free language but L_2 is a context-free language.
(D)	Neither L_1 nor L_2 are context-free languages.

Q.46	Which of the following statement(s) is/are TRUE while computing *First* and *Follow* during top down parsing by a compiler?
(A)	For a production $A \rightarrow \epsilon$, ϵ will be added to $First(A)$.
(B)	If there is any input right end marker, it will be added to $First(S)$, where S is the start symbol.
(C)	For a production $A \rightarrow \epsilon$, ϵ will be added to $Follow(A)$.
(D)	If there is any input right end marker, it will be added to $Follow(S)$, where S is the start symbol.

Q.47	Consider a relational schema $team(name, city, owner)$, with functional dependencies $\{name \rightarrow city,\ name \rightarrow owner\}$. The relation $team$ is decomposed into two relations, $t1(name, city)$ and $t2(name, owner)$. Which of the following statement(s) is/are TRUE?
(A)	The relation $team$ is NOT in BCNF.
(B)	The relations $t1$ and $t2$ are in BCNF.
(C)	The decomposition constitutes a lossless join.
(D)	The relation $team$ is NOT in 3NF.

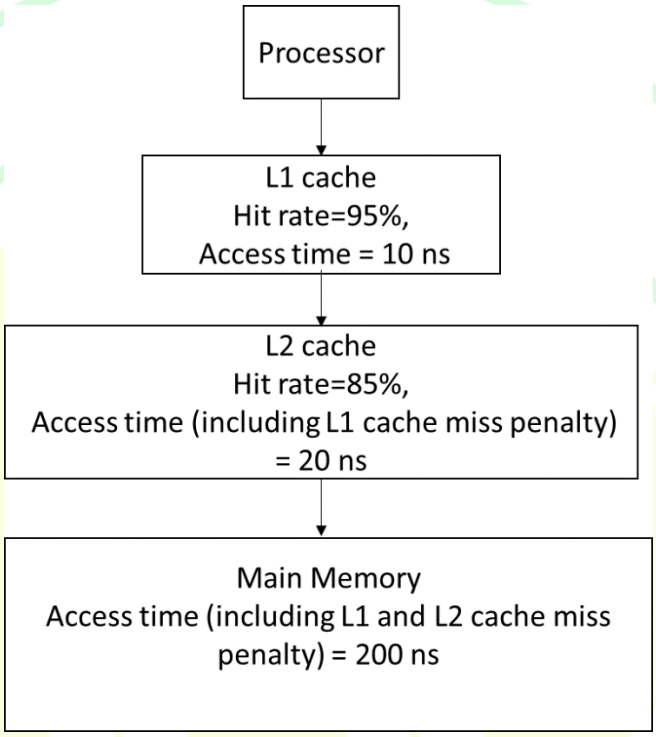
Q.48	Which of the following predicate logic formulae/formula is/are CORRECT representation(s) of the statement: "*Everyone has exactly one mother*"? The meanings of the predicates used are: • $mother(y, x)$: y is the mother of x • $noteq(x, y)$: x and y are not equal
(A)	$\forall x \exists y \exists z \big(mother(y, x) \wedge \neg mother(z, x)\big)$
(B)	$\forall x \exists y \big[mother(y, x) \wedge \forall z \big(noteq(z, y) \rightarrow \neg mother(z, x)\big)\big]$
(C)	$\forall x \forall y [mother(y, x) \rightarrow \exists z (mother(z, x) \wedge \neg noteq(z, y))]$
(D)	$\forall x \exists y \big[mother(y, x) \wedge \neg \exists z \big(noteq(z, y) \wedge mother(z, x)\big)\big]$

Q.49	$A = \{0, 1, 2, 3, \ldots\}$ is the set of non-negative integers. Let F be the set of functions from A to itself. For any two functions, $f_1, f_2 \in$ F, we define $$(f_1 \odot f_2)(n) = f_1(n) + f_2(n)$$ for every number n in A. Which of the following is/are CORRECT about the mathematical structure $(\text{F}, \odot)$?
(A)	$(\text{F}, \odot)$ is an Abelian group.
(B)	$(\text{F}, \odot)$ is an Abelian monoid.
(C)	$(\text{F}, \odot)$ is a non-Abelian group.
(D)	$(\text{F}, \odot)$ is a non-Abelian monoid.

Q.50	Consider the following deterministic finite automaton (DFA) defined over the alphabet, $\Sigma = \{a, b\}$. Identify which of the following language(s) is/are accepted by the given DFA.
	
(A)	The set of all strings containing an even number of b's.
(B)	The set of all strings containing the pattern bab.
(C)	The set of all strings ending with the pattern bab.
(D)	The set of all strings not containing the pattern aba.

Q.51	A disk of size 512M bytes is divided into blocks of 64K bytes. A file is stored in the disk using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a *NULL* pointer (also of 4 bytes). Suppose a file of 1M bytes needs to be stored in the disk. Assume, 1K = 2^{10} and 1M = 2^{20}. The amount of space in bytes that will be wasted due to internal fragmentation is _____. (Answer in integer)

Q.52	Refer to the given 3-address code sequence. This code sequence is split into basic blocks. The number of basic blocks is _____. (Answer in integer)
	1001: i = 1 1002: j = 1 1003: t1 = 10*i 1004: t2 = t1+j 1005: t3 = 8*t2 1006: t4 = t3-88 1007: a[t4] = 0.0 1008: j = j+1 1009: if j <= 10 goto 1003 1010: i = i+1 1011: if i <= 10 goto 1002 1012: i = 1 1013: t5 = i-1 1014: t6 = 88*t5 1015: a[t6] = 1.0 1016: i = i+1 1017: if i <= 10 goto 1013

Q.53	A computer has a memory hierarchy consisting of two-level cache (L1 and L2) and a main memory. If the processor needs to access data from memory, it first looks into L1 cache. If the data is not found in L1 cache, it goes to L2 cache. If it fails to get the data from L2 cache, it goes to main memory, where the data is definitely available. *Hit rates* and *access times* of various memory units are shown in the figure. The *average memory access time* in nanoseconds (*ns*) is _____. (rounded off to two decimal places)
	

Q.54	In optimal page replacement algorithm, information about all future page references is available to the operating system (OS). A modification of the optimal page replacement algorithm is as follows: *The OS correctly predicts only up to next 4 page references (including the current page) at the time of allocating a frame to a page.* A process accesses the pages in the following order of page numbers: $1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1, 4.$ If the system has three memory frames that are initially empty, the number of page faults that will occur during execution of the process is _____ . (Answer in integer)

Q.55 Consider the following database tables of a sports league.

```
player(pid,pname,age)          team(tid,tname,city,cid)
coach(cid,cname)               members(pid,tid)
```

An instance of the table and an SQL query are given.

player

pid	pname	age
1	Jasprit	31
2	Atharva	24
3	Ishan	26
4	Axar	30

coach

cid	cname
101	Ricky
102	Mark
103	Trevor

team

tid	tname	city	cid
10	MI	Mumbai	102
20	DC	Delhi	101
30	PK	Mohali	103

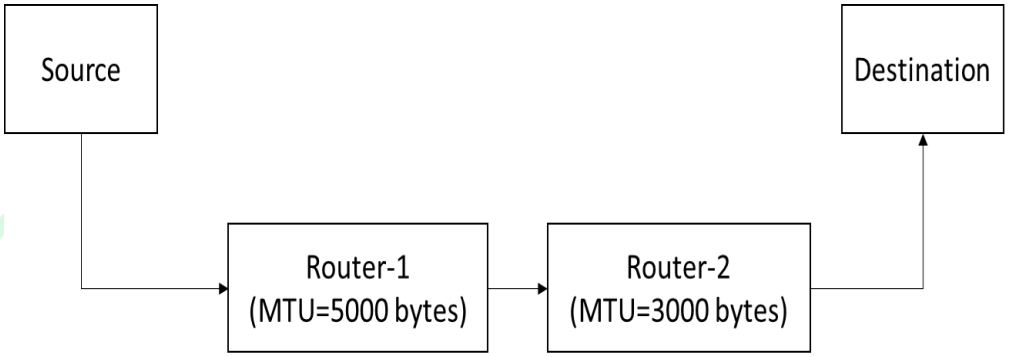
members

pid	tid
1	10
2	30
3	10
4	20

```
SELECT MIN(P.age)
FROM player P
WHERE P.pid IN (
    SELECT M.pid
    FROM team T, coach C, members M
    WHERE C.cname = 'Mark'
          AND T.cid = C.cid
          AND M.tid = T.tid
    )
```

The value returned by the given SQL query is _____ . (Answer in integer)

Q.56	Suppose a 5-bit message is transmitted from a source to a destination through a noisy channel. The probability that a bit of the message gets flipped during transmission is 0.01. Flipping of each bit is independent of one another. The probability that the message is delivered error-free to the destination is _____ . (rounded off to three decimal places)

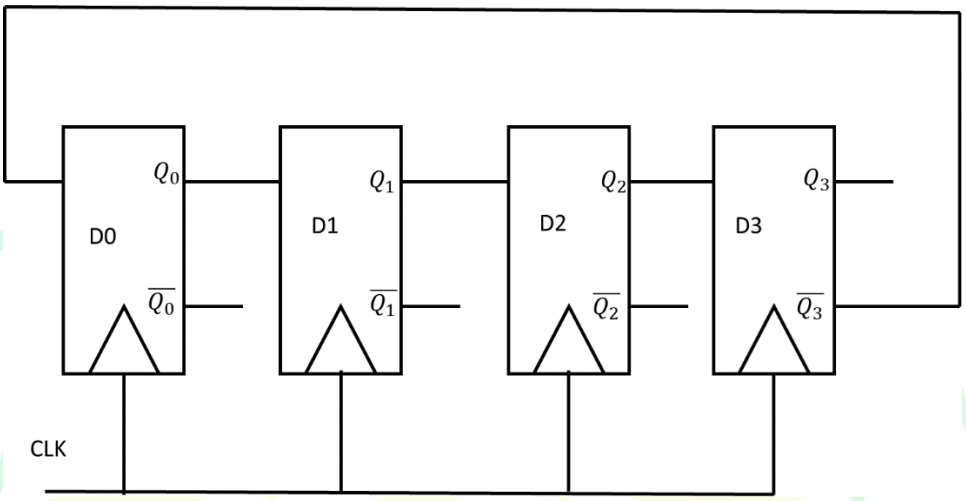
Q.57	Suppose a message of size 15000 bytes is transmitted from a source to a destination using IPv4 protocol via two routers as shown in the figure. Each router has a defined maximum transmission unit (MTU) as shown in the figure, including IP header. The number of fragments that will be delivered to the destination is _____ . (Answer in integer)
	

Q.58	Consider a probability distribution given by the density function $P(x)$. $$P(x) = \begin{cases} Cx^2, & \text{for } 1 \leq x \leq 4 \\ 0, & \text{for } x < 1 \text{ or } x > 4 \end{cases}$$ The probability that x lies between 2 and 3, i.e., $P(2 \leq x \leq 3)$ is _____. (rounded off to three decimal places)

Q.59

Consider a finite state machine (FSM) with one input X and one output f, represented by the given state transition table. The minimum number of states required to realize this FSM is _____. (Answer in integer)

Present state	Next state		Output f	
	$X = 0$	$X = 1$	$X = 0$	$X = 1$
A	F	B	0	0
B	D	C	0	0
C	F	E	0	0
D	G	A	1	0
E	D	C	0	0
F	F	B	1	1
G	G	H	0	1
H	G	A	1	0

Q.60	Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial state). The number of distinct states the circuit will go through before returning back to the initial state is _____ . (Answer in integer)
	

Q.61

```
#include <stdio.h>
int foo(int S[],int size){
    if(size == 0) return 0;
    if(size == 1) return 1;
    if(S[0] != S[1]) return 1+foo(S+1,size-1);
    return foo(S+1,size-1);
}
int main(){
    int A[]={0,1,2,2,2,0,0,1,1};
    printf("%d",foo(A,9));
    return 0;
}
```

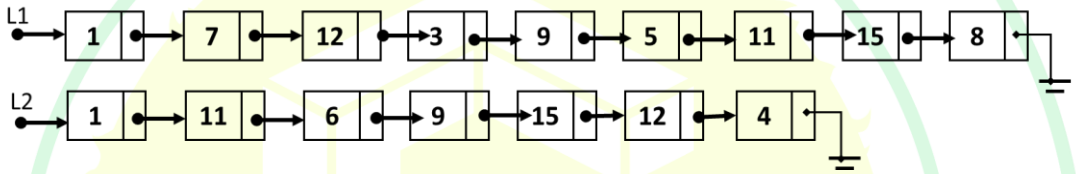
The value printed by the given C program is _____ . (Answer in integer)

Q.62 Let **LIST** be a datatype for an implementation of linked list defined as follows:

```
typedef struct list {
    int data;
    struct list *next;
} LIST;
```

Suppose a program has created two linked lists, L1 and L2, whose contents are given in the figure below (code for creating L1 and L2 is not provided here). L1 contains 9 nodes, and L2 contains 7 nodes.

Consider the following C program segment that modifies the list L1. The number of nodes that will be there in L1 after the execution of the code segment is _____ . (Answer in integer)




```
int find (int query, LIST *list) {
    while (list != NULL){
        if(list->data == query) return 1;
        list = list->next;
    }
    return 0;
}
int main () {
     … … …
    ptr1=L1; ptr2=L2;
    while (ptr1->next != NULL){
        query = ptr1->next->data;
        if (find (query, L2))
            ptr1->next = ptr1->next->next;
        else ptr1 = ptr1->next;
    }
     … … …
    return 0;
}
```

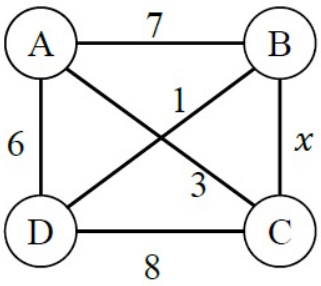
Q.63

Consider the following C program:

```
#include <stdio.h>
int gate (int n) {
    int d, t, newnum, turn;
    newnum = turn = 0; t=1;
    while (n>=t) t *= 10;
    t /=10;
    while (t>0) {
        d = n/t;
        n = n%t;
        t /= 10;
        if (turn) newnum = 10*newnum + d;
        turn = (turn + 1) %2;
    }
    return newnum;
}
int main () {
    printf ("%d", gate(14362));
    return 0;

}
```

The value printed by the given C program is _____ . (Answer in integer)

Q.64	The maximum value of x such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is _____ . (answer in integer)
	

Q.65	In a double hashing scheme, $h_1(k) = k \bmod 11$ and $h_2(k) = 1 + (k \bmod 7)$ are the auxiliary hash functions. The size m of the hash table is 11. The hash function for the i-th probe in the open address table is $[h_1(k) + i\, h_2(k)] \bmod m$. The following keys are inserted in the given order: 63, 50, 25, 79, 67, 24. The slot at which key 24 gets stored is _____. (Answer in integer)

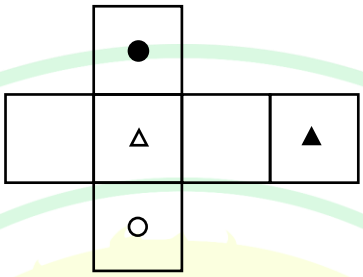
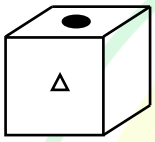
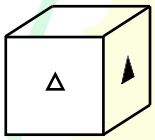
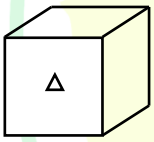
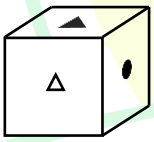
General Aptitude

Q.1 – Q.5 Carry ONE mark Each

Q.1	Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered. Select the most appropriate option to complete the above sentence.
(A)	ambivalence
(B)	satisfaction
(C)	resolve
(D)	revolve

Q.2	Bird : Nest :: Bee : _____ Select the correct option to complete the analogy.
(A)	Kennel
(B)	Hammock
(C)	Hive
(D)	Lair

Q.3	If $Pe^{x} = Qe^{-x}$ for all real values of x, which one of the following statements is true?
(A)	$P = Q = 0$
(B)	$P = Q = 1$
(C)	$P = 1;\ Q = -1$
(D)	$\frac{P}{Q} = 0$

Q.4	The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube? Note: The figures shown are representative.
	
(A)	
(B)	
(C)	
(D)	

Q.5	Let p_1 and p_2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of p_1 and p_2?
(A)	$p_1 + p_2$ is not a prime number.
(B)	$p_1 p_2$ is not a prime number.
(C)	$p_1 + p_2 + 1$ is a prime number.
(D)	$p_1 p_2 + 1$ is a prime number.

Q.6 – Q.10 Carry TWO marks Each

Q.6	Based only on the conversation below, identify the logically correct inference: “Even if I had known that you were in the hospital, I would not have gone there to see you”, Ramya told Josephine.
(A)	Ramya knew that Josephine was in the hospital.
(B)	Ramya did not know that Josephine was in the hospital.
(C)	Ramya and Josephine were once close friends; but now, they are not.
(D)	Josephine was in the hospital due to an injury to her leg.

Q.7	If IMAGE and FIELD are coded as FHBNJ and EMFJG respectively then, which one among the given options is the most appropriate code for BEACH ?
(A)	CEADP
(B)	IDBFC
(C)	JGIBC
(D)	IBCEC

Q.8 Which one of the following options is correct for the given data in the table?

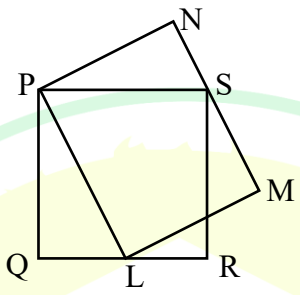
Iteration (i)	0	1	2	3
Input (I)	20	-4	10	15
Output (X)	20	16	26	41
Output (Y)	20	-80	-800	-12000

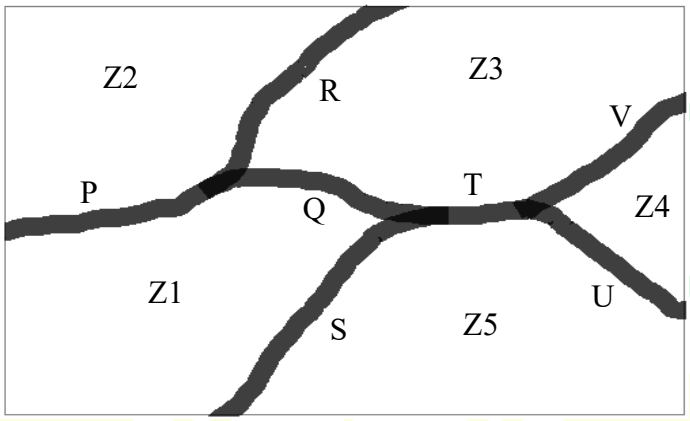
(A) $X(i) = X(i-1) + I(i); \quad Y(i) = Y(i-1)I(i); \quad i > 0$

(B) $X(i) = X(i-1)I(i); \quad Y(i) = Y(i-1) + I(i); \quad i > 0$

(C) $X(i) = X(i-1)I(i); \quad Y(i) = Y(i-1)I(i); \quad i > 0$

(D) $X(i) = X(i-1) + I(i); \quad Y(i) = Y(i-1)I(i-1); \quad i > 0$

Q.9	In the given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of the rectangle is on the side QR. Side MN of the rectangle passes through the corner S of the square. What is the area (in cm^2) of the rectangle PLMN? Note: The figure shown is representative.
	
(A)	$2\sqrt{2}$
(B)	2
(C)	8
(D)	4

Q.10	The diagram below shows a river system consisting of 7 segments, marked P, Q, R, S, T, U, and V. It splits the land into 5 zones, marked Z1, Z2, Z3, Z4, and Z5. We need to connect these zones using the least number of bridges. Out of the following options, which one is correct? Note: The figure shown is representative.
	
(A)	Bridges on P, Q, and T
(B)	Bridges on P, Q, S, and T
(C)	Bridges on Q, R, T, and V
(D)	Bridges on P, Q, S, U, and V

Q.11 – Q.35 Carry ONE mark Each

Q.11	If $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$, then which ONE of the following is A^8 ?
(A)	$\begin{pmatrix} 25 & 0 \\ 0 & 25 \end{pmatrix}$
(B)	$\begin{pmatrix} 125 & 0 \\ 0 & 125 \end{pmatrix}$
(C)	$\begin{pmatrix} 625 & 0 \\ 0 & 625 \end{pmatrix}$
(D)	$\begin{pmatrix} 3125 & 0 \\ 0 & 3125 \end{pmatrix}$
Q.12	The value of x such that $x > 1$, satisfying the equation $\int_1^x t \ln t \, dt = \frac{1}{4}$ is
(A)	$\sqrt{e}$
(B)	e
(C)	e^2
(D)	$e - 1$

Q.13	Consider a binary tree T in which every node has either zero or two children. Let $n > 0$ be the number of nodes in T. Which ONE of the following is the number of nodes in T that have exactly two children?
(A)	$\frac{n-2}{2}$
(B)	$\frac{n-1}{2}$
(C)	$\frac{n}{2}$
(D)	$\frac{n+1}{2}$

Q.14	Let $L, M,$ and N be non-singular matrices of order 3 satisfying the equations $L^2 = L^{-1}$, $M = L^8$ and $N = L^2$. Which ONE of the following is the value of the determinant of $(M - N)$?
(A)	0
(B)	1
(C)	2
(D)	3
Q.15	Let $P(x)$ be an arbitrary predicate over the domain of natural numbers. Which ONE of the following statements is TRUE?
(A)	$\big(P(0) \wedge (\forall x\,[P(x) \Rightarrow P(x+1)])\big) \Rightarrow \big(\forall x\,P(x)\big)$
(B)	$\big(P(0) \wedge (\forall x\,[P(x) \Rightarrow P(x-1)])\big) \Rightarrow \big(\forall x\,P(x)\big)$
(C)	$\big(P(1000) \wedge (\forall x\,[P(x) \Rightarrow P(x-1)])\big) \Rightarrow \big(\forall x\,P(x)\big)$
(D)	$\big(P(1000) \wedge (\forall x\,[P(x) \Rightarrow P(x+1)])\big) \Rightarrow \big(\forall x\,P(x)\big)$

Q.16	Consider the following statements: (i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address. (ii) A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S. Which ONE of the following is CORRECT?
(A)	Both (i) and (ii) are TRUE
(B)	(i) is TRUE and (ii) is FALSE
(C)	(i) is FALSE and (ii) is TRUE
(D)	Both (i) and (ii) are FALSE

<table>
<tr><td>Q.17</td><td>Consider the routing protocols given in List I and the names given in List II:
<table><tr><th colspan="2">List I</th><th colspan="2">List II</th></tr><tr><td>(i)</td><td>Distance vector routing</td><td>(a)</td><td>Bellman-Ford</td></tr><tr><td>(ii)</td><td>Link state routing</td><td>(b)</td><td>Dijkstra</td></tr></table>For matching of items in List I with those in List II, which ONE of the following options is CORRECT?</td></tr>
<tr><td></td><td></td></tr>
<tr><td>(A)</td><td>(i) – (a) and (ii) – (b)</td></tr>
<tr><td>(B)</td><td>(i) – (a) and (ii) – (a)</td></tr>
<tr><td>(C)</td><td>(i) – (b) and (ii) – (a)</td></tr>
<tr><td>(D)</td><td>(i) – (b) and (ii) – (b)</td></tr>
<tr><td></td><td></td></tr>
</table>

Q.18	A machine receives an IPv4 datagram. The protocol field of the IPv4 header has the protocol number of a protocol X. Which ONE of the following is NOT a possible candidate for X?
(A)	Internet Control Message Protocol (ICMP)
(B)	Internet Group Management Protocol (IGMP)
(C)	Open Shortest Path First (OSPF)
(D)	Routing Information Protocol (RIP)

Q.19 Consider the following C program:

```
#include <stdio.h>
void stringcopy(char *, char *);
int main(){
      char a[30] = "@#Hello World!";
      stringcopy(a, a + 2);
      printf("%s\n", a);
      return 0;
}
void stringcopy(char *s, char *t) {
      while(*t)
            *s++ = *t++;
}
```

Which ONE of the following will be the output of the program?

(A) `@#Hello World!`

(B) `Hello World!`

(C) `ello World!`

(D) `Hello World!d!`

Q.20	Consider an unordered list of N distinct integers. What is the minimum number of element comparisons required to find an integer in the list that is NOT the largest in the list?
(A)	1
(B)	$N-1$
(C)	N
(D)	$2N-1$
Q.21	Consider the following statements about the use of backpatching in a compiler for intermediate code generation: (I) Backpatching can be used to generate code for Boolean expression in one pass. (II) Backpatching can be used to generate code for flow-of-control statements in one pass. Which ONE of the following options is CORRECT?
(A)	Only (I) is correct.
(B)	Only (II) is correct.
(C)	Both (I) and (II) are correct.
(D)	Neither (I) nor (II) is correct.

Q.22	Given the following syntax directed translation rules: Rule 1: $R \rightarrow AB \{B.i = R.i - 1; A.i = B.i; R.i = A.i + 1;\}$ Rule 2: $P \rightarrow CD \{P.i = C.i + D.i; D.i = C.i + 2;\}$ Rule 3: $Q \rightarrow EF \{Q.i = E.i + F.i;\}$ Which ONE is the CORRECT option among the following?
(A)	Rule 1 is S-attributed and L-attributed; Rule 2 is S-attributed and not L-attributed; Rule 3 is neither S-attributed nor L-attributed
(B)	Rule 1 is neither S-attributed nor L-attributed; Rule 2 is S-attributed and L-attributed; Rule 3 is S-attributed and L-attributed
(C)	Rule 1 is neither S-attributed nor L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed
(D)	Rule 1 is S-attributed and not L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed

Q.23	Consider a network that uses Ethernet and IPv4. Assume that IPv4 headers do not use any options field. Each Ethernet frame can carry a maximum of 1500 bytes in its data field. A UDP segment is transmitted. The payload (data) in the UDP segment is 7488 bytes. Which ONE of the following choices has the CORRECT total number of fragments transmitted and the size of the last fragment including IPv4 header?
(A)	5 fragments, 1488 bytes
(B)	6 fragments, 88 bytes
(C)	6 fragments, 108 bytes
(D)	6 fragments, 116 bytes
Q.24	Which ONE of the following languages is accepted by a deterministic pushdown automaton?
(A)	Any regular language.
(B)	Any context-free language.
(C)	Any language accepted by a non-deterministic pushdown automaton.
(D)	Any decidable language.

Q.25	Let G_1, G_2 be Context Free Grammars (CFGs) and R be a regular expression. For a grammar G, let $L(G)$ denote the language generated by G. Which ONE among the following questions is decidable?
(A)	Is $L(G_1) = L(G_2)$?
(B)	Is $L(G_1) \cap L(G_2) = \emptyset$?
(C)	Is $L(G_1) = L(R)$?
(D)	Is $L(G_1) = \emptyset$?
Q.26	Processes $P1$, $P2$, $P3$, $P4$ arrive in that order at times 0, 1, 2, and 8 milliseconds respectively, and have execution times of 10, 13, 6, and 9 milliseconds respectively. Shortest Remaining Time First (SRTF) algorithm is used as the CPU scheduling policy. Ignore context switching times. Which ONE of the following correctly gives the average turnaround time of the four processes in milliseconds?
(A)	22
(B)	15
(C)	37
(D)	19

Q.27	An audit of a banking transactions system has found that on an earlier occasion, two joint holders of account A attempted simultaneous transfers of Rs. 10000 each from account A to account B. Both transactions read the same value, Rs. 11000, as the initial balance in A and were allowed to go through. B was credited Rs. 10000 twice. A was debited only once and ended up with a balance of Rs. 1000. Which of the following properties is/are certain to have been violated by the system?
(A)	Atomicity
(B)	Consistency
(C)	Isolation
(D)	Durability
Q.28	Which of the following is/are part of an Instruction Set Architecture of a processor?
(A)	The size of the cache memory
(B)	The clock frequency of the processor
(C)	The number of cache memory levels
(D)	The total number of registers

Q.29	Which of the following statements regarding Breadth First Search (BFS) and Depth First Search (DFS) on an undirected simple graph G is/are TRUE?
(A)	A DFS tree of G is a Shortest Path tree of G.
(B)	Every non-tree edge of G with respect to a DFS tree is a forward/back edge.
(C)	If (u, v) is a non-tree edge of G with respect to a BFS tree, then the distances from the source vertex s to u and v in the BFS tree are within ± 1 of each other.
(D)	Both BFS and DFS can be used to find the connected components of G.

Q.30 Consider the two lists **List I** and **List II** given below:

List I	List II
(i) Context free languages	(a) Closed under union
(ii) Recursive languages	(b) Not closed under complementation
(iii) Regular languages	(c) Closed under intersection

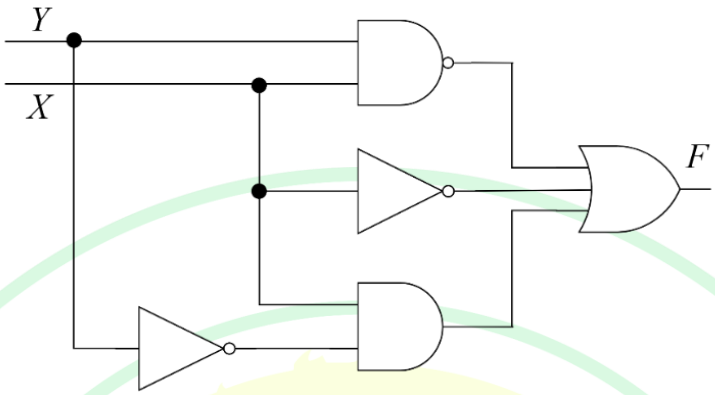
For matching of items in **List I** with those in **List II,** which of the following option(s) is/are CORRECT?

(A) (i) – (a), (ii) – (b), and (iii) – (c)

(B) (i) – (b), (ii) – (a), and (iii) – (c)

(C) (i) – (b), (ii) – (c), and (iii) – (a)

(D) (i) – (a), (ii) – (c), and (iii) – (b)

Q.31	Consider the following logic circuit diagram.  Which is/are the CORRECT option(s) for the output function F?
(A)	$\overline{X\,Y}$
(B)	$\bar{X} + \bar{Y} + X\bar{Y}$
(C)	$\overline{X\,Y} + \bar{X} + X\bar{Y}$
(D)	$X + \bar{Y}$
Q.32	The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's algorithm: M: 1100 1101 1110 1101 and Q: 1010 0100 1010 1010 The total number of addition and subtraction operations to be performed is _____. (Answer in integer)

Q.33	```
	int x=126,y=105;
	do {
	if(x>y) x=x-y;
	else y=y-x;
	} while(x!=y);
	
	printf("%d",x);
	``
	The output of the given C code segment is _____. (Answer in integer)

Q.34	In a 4-bit ripple counter, if the period of the waveform at the last flip-flop is 64 microseconds, then the frequency of the ripple counter in kHz is _____. (Answer in integer)
Q.35	Suppose the values $10, -4, 15, 30, 20, 5, 60, 19$ are inserted in that order into an initially empty binary search tree. Let T be the resulting binary search tree. The number of edges in the path from the node containing 19 to the root node of T is _____. (Answer in integer)

Q.36 – Q.65 Carry TWO marks Each

Q.36	Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps (bits/second) and the propagation delay between the two nodes is 100 milliseconds. Assume that the processing times at the source and destination are negligible. Also, assume that the size of the acknowledgement packet is negligible. Which ONE of the following most accurately gives the channel utilization for the above scenario in percentage?
(A)	88.23
(B)	93.75
(C)	85.44
(D)	66.67

Q.37	Let G be an edge-weighted undirected graph with positive edge weights. Suppose a positive constant α is added to the weight of every edge. Which ONE of the following statements is TRUE about the minimum spanning trees (MSTs) and shortest paths (SPs) in G before and after the edge weight update?
(A)	Every MST remains an MST, and every SP remains an SP.
(B)	MSTs need not remain MSTs, and every SP remains an SP.
(C)	Every MST remains an MST, and SPs need not remain SPs.
(D)	MSTs need not remain MSTs, and SPs need not remain SPs.

Q.38	A meld operation on two instances of a data structure combines them into one single instance of the same data structure. Consider the following data structures: P: Unsorted doubly linked list with pointers to the head node and tail node of the list. Q: Min-heap implemented using an array. R: Binary Search Tree. Which ONE of the following options gives the worst-case time complexities for meld operation on instances of size n of these data structures?
(A)	P: $\Theta(1)$, Q: $\Theta(n)$, R: $\Theta(n)$
(B)	P: $\Theta(1)$, Q: $\Theta(n \log n)$, R: $\Theta(n)$
(C)	P: $\Theta(n)$, Q: $\Theta(n \log n)$, R: $\Theta(n^2)$
(D)	P: $\Theta(1)$, Q: $\Theta(n)$, R: $\Theta(n \log n)$

Q.39	For a direct-mapped cache, 4 bits are used for the tag field and 12 bits are used to index into a cache block. The size of each cache block is one byte. Assume that there is no other information stored for each cache block. Which ONE of the following is the CORRECT option for the sizes of the main memory and the cache memory in this system (byte addressable), respectively?
(A)	64 KB and 4 KB
(B)	128 KB and 16 KB
(C)	64 KB and 8 KB
(D)	128 KB and 6 KB
Q.40	Given a Context-Free Grammar G as follows: $S \rightarrow Aa \mid bAc \mid dc \mid bda$ $A \rightarrow d$ Which ONE of the following statements is TRUE?
(A)	G is neither LALR(1) nor SLR(1)
(B)	G is CLR(1), not LALR(1)
(C)	G is LALR(1), not SLR(1)
(D)	G is LALR(1), also SLR(1)

Q.41	An array A of length n with distinct elements is said to be bitonic if there is an index $1 \leq i \leq n$ such that $A[1..i]$ is sorted in the non-decreasing order and $A[i+1..n]$ is sorted in the non-increasing order. Which ONE of the following represents the best possible asymptotic bound for the worst-case number of comparisons by an algorithm that searches for an element in a bitonic array A?
(A)	$\Theta(n)$
(B)	$\Theta(1)$
(C)	$\Theta(\log^2 n)$
(D)	$\Theta(\log n)$
Q.42	Let $\mathcal{F}$ be the set of all functions from $\{1, \ldots, n\}$ to $\{0,1\}$. Define the binary relation $\preccurlyeq$ on $\mathcal{F}$ as follows: $\forall f, g \in \mathcal{F}$, $f \preccurlyeq g$ if and only if $\forall x \in \{1, \ldots, n\}$, $f(x) \leq g(x)$, where $0 \leq 1$. Which of the following statement(s) is/are TRUE?
(A)	$\preccurlyeq$ is a symmetric relation
(B)	$(\mathcal{F}, \preccurlyeq)$ is a partial order
(C)	$(\mathcal{F}, \preccurlyeq)$ is a lattice
(D)	$\preccurlyeq$ is an equivalence relation

Q.43	Given the following Karnaugh Map for a Boolean function $F(w, x, y, z)$: yz / wx 1 0 0 1 0 1 1 0 0 1 1 0 1 0 0 1 Which one or more of the following Boolean expression(s) represent(s) F?
(A)	$\bar{w}\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + w\bar{x}yz + xz$
(B)	$\bar{w}\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + w\bar{x}yz + xz$
(C)	$\bar{w}\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}z + xz$
(D)	$\bar{x}\bar{z} + xz$

Q.44	Consider a system of linear equations $PX = Q$ where $P \in \mathbb{R}^{3\times3}$ and Q $\in \mathbb{R}^{3\times1}$. Suppose P has an LU decomposition, $P = LU$, where $L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$ and $U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$. Which of the following statement(s) is/are TRUE?
(A)	The system $PX = Q$ can be solved by first solving $LY = Q$ and then $UX = Y$.
(B)	If P is invertible, then both L and U are invertible.
(C)	If P is singular, then at least one of the diagonal elements of U is zero.
(D)	If P is symmetric, then both L and U are symmetric.

Q.45	Consider a stack data structure into which we can PUSH and POP records. Assume that each record pushed in the stack has a positive integer key and that all keys are distinct. We wish to augment the stack data structure with an $O(1)$ time MIN operation that returns a pointer to the record with smallest key present in the stack 1) without deleting the corresponding record, and 2) without increasing the complexities of the standard stack operations. Which one or more of the following approach(es) can achieve it?
(A)	Keep with every record in the stack, a pointer to the record with the smallest key below it.
(B)	Keep a pointer to the record with the smallest key in the stack.
(C)	Keep an auxiliary array in which the key values of the records in the stack are maintained in sorted order.
(D)	Keep a Min-Heap in which the key values of the records in the stack are maintained.

Q.46	Consider the following relational schema along with all the functional dependencies that hold on them. *R1(A, B, C, D, E):* $\{D \rightarrow E,\ EA \rightarrow B,\ EB \rightarrow C\}$ *R2(A, B, C, D):* $\{A \rightarrow D,\ A \rightarrow B,\ C \rightarrow A\}$ Which of the following statement(s) is/are TRUE?
(A)	*R1* is in 3NF
(B)	*R2* is in 3NF
(C)	*R1* is NOT in 3NF
(D)	*R2* is NOT in 3NF

Q.47

Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 1 6 4 5 1 3 2. The contents of the frames are as follows initially and after each reference (from left to right):

initially	*after*												
-	1*	2*	3*	4*	5*	4	1	6*	4	5	1*	3*	2*
-	1	1	1	1	1	1	1	6	6	6	6	6	2
-	-	2	2	4	4	4	4	4	4	4	1	1	1
-	-	-	3	3	5	5	5	5	5	5	5	3	3

The *-marked references cause page replacements.

Which one or more of the following could be the page replacement policy/policies in use?

(A) Least Recently Used page replacement policy

(B) Least Frequently Used page replacement policy

(C) Most Frequently Used page replacement policy

(D) Optimal page replacement policy

Q.48	$P = \{P_1, P_2, P_3, P_4\}$ consists of all active processes in an operating system. $R = \{R_1, R_2, R_3, R_4\}$ consists of single instances of distinct types of resources in the system. The resource allocation graph has the following assignment and claim edges. Assignment edges: $R_1 \rightarrow P_1$, $R_2 \rightarrow P_2$, $R_3 \rightarrow P_3$, $R_4 \rightarrow P_4$ (the assignment edge $R_1 \rightarrow P_1$ means resource R_1 is assigned to process P_1 , and so on for others) Claim edges: $P_1 \rightarrow R_2$, $P_2 \rightarrow R_3$, $P_3 \rightarrow R_1$, $P_2 \rightarrow R_4$, $P_4 \rightarrow R_2$ (the claim edge $P_1 \rightarrow R_2$ means process P_1 is waiting for resource R_2 , and so on for others) Which of the following statement(s) is/are CORRECT?
(A)	Aborting P_1 makes the system deadlock free.
(B)	Aborting P_3 makes the system deadlock free.
(C)	Aborting P_2 makes the system deadlock free.
(D)	Aborting P_1 and P_4 makes the system deadlock free.

Q.49	Three floating point numbers X, Y, and Z are stored in three registers R_X, R_Y, and R_Z, respectively in IEEE 754 single precision format as given below in hexadecimal: R_X = 0xC1100000, R_Y = 0x40C00000, and R_Z = 0x41400000 Which of the following option(s) is/are CORRECT?
(A)	$4(X + Y) + Z = 0$
(B)	$2Y - Z = 0$
(C)	$4X + 3Z = 0$
(D)	$X + Y + Z = 0$
Q.50	Which of the following Boolean algebraic equation(s) is/are CORRECT?
(A)	$\bar{A}\,B\,C + A\,\bar{B}\,\bar{C} + \bar{A}\,\bar{B}\,\bar{C} + A\,\bar{B}\,C + A\,B\,C = B\,C + \bar{B}\,\bar{C} + \bar{A}\,\bar{B}$
(B)	$A\,B + \bar{A}\,C + B\,C = A\,B + \bar{A}\,C$
(C)	$(A + C)\,(\bar{A} + B) = A\,B + \bar{A}\,C$
(D)	$\overline{(A + \bar{B} + \bar{D})\,(C + D)\,(\bar{A} + C + D)\,(A + B + \bar{D})} = \bar{A}\,D + \bar{C}\,\bar{D}$

Q.51	Consider two grammars G_1 and G_2 with the production rules given below: $G_1: S \rightarrow if\ \ E\ then\ \ S \mid if\ E\ then\ S\ else\ S \mid a$ $E \rightarrow b$ $G_2: S \rightarrow if\ E\ then\ S \mid M$ $M \rightarrow if\ E\ then\ M\ else\ S \mid c$ $E \rightarrow b$ where $if, then, else, a, b, c$ are the terminals. Which of the following option(s) is/are CORRECT?
(A)	G_1 is not $LL(1)$ and G_2 is $LL(1)$.
(B)	G_1 is $LL(1)$ and G_2 is not $LL(1)$.
(C)	G_1 and G_2 are not $LL(1)$.
(D)	G_1 and G_2 are ambiguous.

Q.52	Let $\Sigma = \{a, b, c\}$. For $x \in \Sigma^*$, and $\alpha \in \Sigma$, let $\#_\alpha(x)$ denote the number of occurrences of α in x. Which one or more of the following option(s) define(s) regular language(s)?
(A)	$\{a^m b^n \mid m, n \geq 0\}$
(B)	$\{a, b\}^* \cap \{a^m b^n c^{m-n} \mid m \geq n \geq 0\}$
(C)	$\{w \mid w \in \{a, b\}^*, \#_a(w) \equiv 2 \pmod 7, \text{and } \#_b(w) \equiv 3 \pmod 9\}$
(D)	$\{w \mid w \in \{a, b\}^*, \#_a(w) \equiv 2 \pmod 7, \text{and } \#_a(w) = \#_b(w)\}$

Q.53	Consider the database transactions T1 and T2, and data items X and Y. Which of the schedule(s) is/are conflict serializable?
	Transaction T1 R1(X) W1(Y) R1(X) W1(X) COMMIT(T1) **Transaction T2** W2(X) W2(Y) COMMIT(T2)
(A)	R1(X), W2(X), W1(Y), W2(Y), R1(X), W1(X), COMMIT(T2), COMMIT(T1)
(B)	W2(X), R1(X), W2(Y), W1(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)
(C)	R1(X), W1(Y), W2(X), W2(Y), R1(X), W1(X), COMMIT(T1), COMMIT(T2)
(D)	W2(X), R1(X), W1(Y), W2(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)

Q.54 Consider the following relational schema:

Students (rollno: integer, name: string, age: integer, cgpa: real)

Courses (courseno: integer, cname: string, credits: integer)

Enrolled (rollno: integer, courseno: integer, grade: string)

Which of the following options is/are correct SQL query/queries to retrieve the names of the students enrolled in course number (i.e., courseno) 1470?

(A)

```
SELECT S.name
FROM Students S
WHERE EXISTS (SELECT * FROM Enrolled E
              WHERE E.courseno = 1470
                    AND E.rollno = S.rollno);
```

(B)

```
SELECT S.name
FROM Students S
WHERE SIZEOF (SELECT * FROM Enrolled E
              WHERE E.courseno = 1470
                    AND E.rollno = S.rollno) > 0;
```

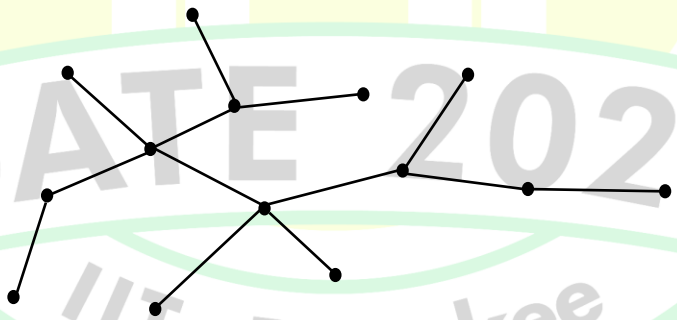
(C)

```
SELECT S.name
FROM Students S
WHERE  0 < (SELECT COUNT(*)
            FROM Enrolled E
            WHERE E.courseno = 1470
                  AND E.rollno = S.rollno);
```

(D)

```
SELECT S.name
FROM Students S NATURAL JOIN Enrolled E
WHERE E.courseno = 1470;
```

Q.55	Given a computing system with two levels of cache (L1 and L2) and a main memory. The first level (L1) cache access time is 1 nanosecond (ns) and the "hit rate" for L1 cache is 90% while the processor is accessing the data from L1 cache. Whereas, for the second level (L2) cache, the "hit rate" is 80% and the "miss penalty" for transferring data from L2 cache to L1 cache is 10 ns. The "miss penalty" for the data to be transferred from main memory to L2 cache is 100 ns. Then the average memory access time in this system in nanoseconds is _____ . (rounded off to one decimal place)
Q.56	A 5-stage instruction pipeline has stage delays of 180, 250, 150, 170, and 250, respectively, in nanoseconds. The delay of an inter-stage latch is 10 nanoseconds. Assume that there are no pipeline stalls due to branches and other hazards. The time taken to process 1000 instructions in microseconds is _____ . (rounded off to two decimal places)
Q.57	In a B^+- tree where each node can hold at most four key values, a root to leaf path consists of the following nodes: A = (49, 77, 83, -), B = (7, 19, 33, 44), C = (20*, 22*, 25*, 26*) The *-marked keys signify that these are data entries in a leaf. Assume that a pointer between keys k_1 and k_2 points to a subtree containing keys in $[k_1, k_2)$, and that when a leaf is created, the smallest key in it is copied up into its parent. A record with key value 23 is inserted into the B^+- tree. The smallest key value in the parent of the leaf that contains 25* is _____. (Answer in integer)

Q.58	A computer system supports a logical address space of 2^{32} bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b-bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size. The value of b is _____ . (Answer in integer)
Q.59	Consider the following algorithm `someAlgo` that takes an undirected graph G as input. `someAlgo(`G`)` 1. Let v be any vertex in G. Run BFS on G starting at v. Let u be a vertex in G at maximum distance from v as given by the BFS. 2. Run BFS on G again with u as the starting vertex. Let z be the vertex at maximum distance from u as given by the BFS. 3. Output the distance between u and z in G. The output of `someAlgo(`T`)` for the tree shown in the given figure is _____. (Answer in integer)
	

Q.60

Let $\Sigma = \{1,2,3,4\}$. For $x \in \Sigma^*$, let $prod(x)$ be the product of symbols in x modulo 7. We take $prod(\epsilon) = 1$, where ϵ is the null string.

For example, $prod(124) = (1 \times 2 \times 4) \bmod 7 = 1$.

Define $L = \{x \in \Sigma^* \mid prod(x) = 2\}$.

The number of states in a minimum state DFA for L is _____. (Answer in integer)

Q.61

An application executes 6.4×10^8 number of instructions in 6.3 seconds. There are four types of instructions, the details of which are given in the table. The duration of a clock cycle in nanoseconds is _____. (rounded off to one decimal place)

Instruction type	Clock cycles required per instruction (CPI)	Number of instructions executed
Branch	2	2.25×10^8
Load	5	1.20×10^8
Store	4	1.65×10^8
Arithmetic	3	1.30×10^8

Q.62

Consider the following C program:

```
#include <stdio.h>
int main(){
      int a;
      int arr[5] = {30,50,10};
      int *ptr;
      ptr = &arr[0] + 1;
      a = *ptr;
      (*ptr)++;
      ptr++;
      printf("%d", a + (*ptr) + arr[1]);
      return 0;
}
```

The output of the above program is _____. (Answer in integer)

Q.63

Consider the following C program:

```
#include <stdio.h>

int g(int n) {
    return (n+10);
}

int f(int n) {
    return g(n*2);
}

int main() {
    int sum, n;
    sum=0;
    for (n=1; n<3; n++)
        sum += g(f(n));
    printf ("%d", sum);
    return 0;

}
```

The output of the given C program is _____. (Answer in integer)

Q.64

A quadratic polynomial $(x-\alpha)(x-\beta)$ over complex numbers is said to be square invariant if $(x-\alpha)(x-\beta) = (x-\alpha^2)(x-\beta^2)$. Suppose from the set of all square invariant quadratic polynomials we choose one at random.

The probability that the roots of the chosen polynomial are equal is _____. (rounded off to one decimal place)

Q.65	The unit interval $(0,1)$ is divided at a point chosen uniformly distributed over $(0,1)$ in $\mathbb{R}$ into two disjoint subintervals. The expected length of the subinterval that contains 0.4 is _____. (rounded off to two decimal places)